OBP FINDER - Analysis Report

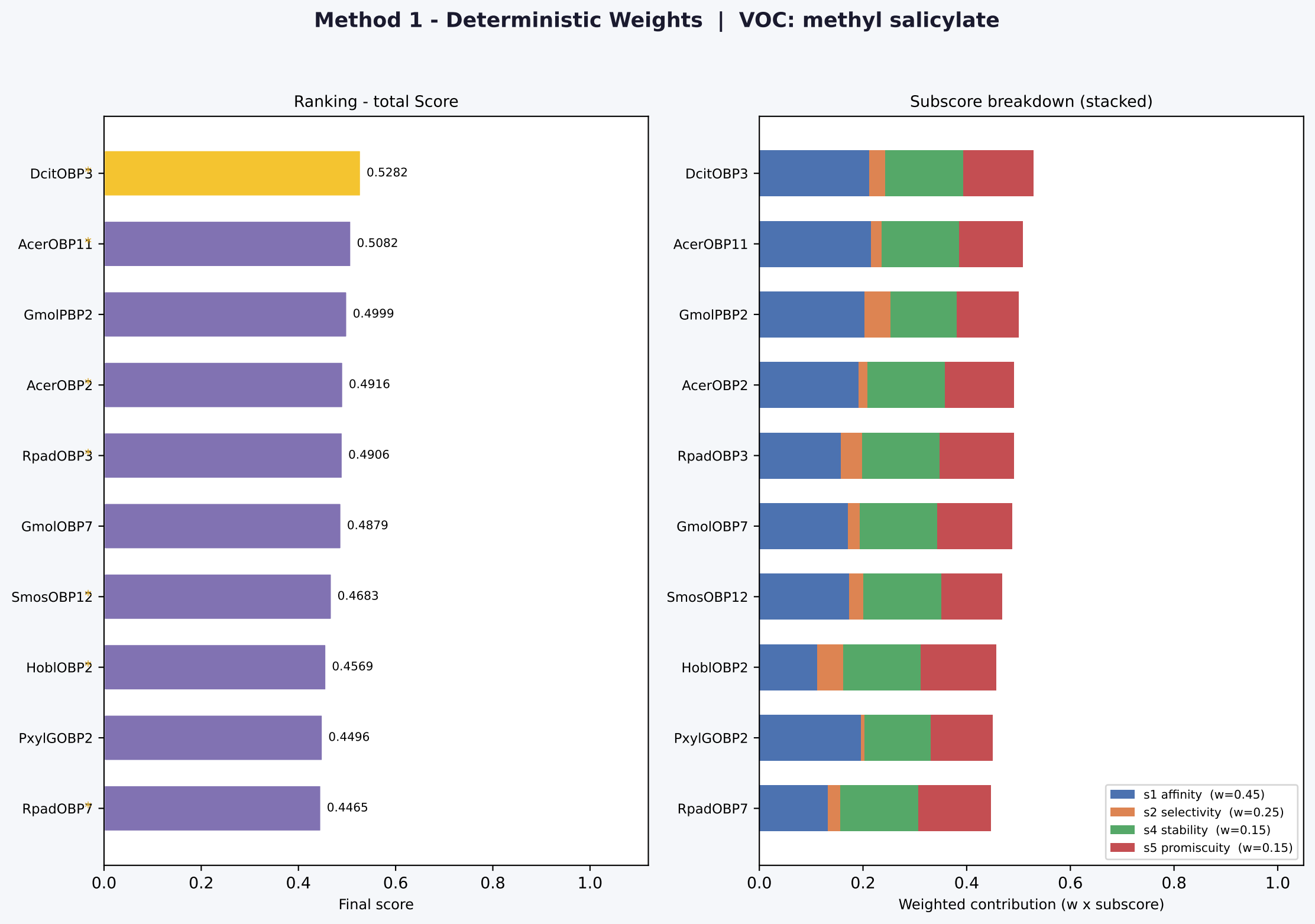
iGEM URV 2025 · Full automatic report (3 methods)

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Target VOC       :  methyl salicylate
CAS              :  119-36-8
Interferents     :  caryophyllene, β-Cyclocitral, diisobutyl phthalate, 10:Ald, hexanal, 1-octen-3-ol, acetic acid, 1-octen-3-one, C8:Ald, C9:Ald, benzaldehyde, 2-hexen-1-ol, 1-hexanol, 2-ethyl-1-hexanol, caproic acid, methyl salicylate, Farnesene, beta-farnesene, nerol, (+, stearic acid, 1-octanol, 1-nonanol, 2-pentylfuran, Ocimene, Geranylacetone, 3-octanone, benzyl alcohol, 2-phenyl ethanol, 2-undecanone, Phellandrene, α-Terpinene, r-limonene, phenylethanal, γ-Terpinene, 12:Ald, furfural, alpha Terpineol, Ethyl hexadecanoate, geraniol, p-xylene, styrene, β-Sesquiphellandrene
Top N candidates :  10

--- Weights ----------------------------
tau (selectivity):  5
w1 affinity      :  0.450
w2 selectivity   :  0.250
w4 stability     :  0.150
w5 promiscuity   :  0.150

--- SMAA configuration -----------------
Method 2 (Perturbation):  c = 50%,  100,000 iterations
Method 3 (pure SMAA)   :  100,000 iterations
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Method 1 - Deterministic Weights | VOC: methyl salicylate
Ranking - total Score
DcitOBP3
AcerOBP11
GmolPBP2
AcerOBP2
RpadOBP3
GmolOBP7
SmosOBP12
HoblOBP2
PxylGOBP2
RpadOBP7
0.5282
0.5082
0.4999
0.4916
0.4906
0.4879
0.4683
0.4569
0.4496
0.4465
0.0
0.2
0.4
0.6
0.8
1.0
Final score
Subscore breakdown (stacked)
s1 affinity (w=0.45)
s2 selectivity (w=0.25)
s4 stability (w=0.15)
s5 promiscuity (w=0.15)
Weighted contribution (w x subscore)

Method 1 - Detailed Table | VOC: methyl salicylate

#	OBP	Score	s1	s2	s4	s5	Ki (uM)	Cys	Pref	Ki interf	Type
1	DcitOBP3	0.5282	0.472	0.122	1.000	0.902	7.01	6	*	4.28	Classic OBP
2	AcerOBP11	0.5082	0.479	0.082	1.000	0.814	6.83	6	*	2.79	Classic OBP
3	GmolPBP2	0.4999	0.452	0.200	0.850	0.793	7.55	7		7.55	PBP
4	AcerOBP2	0.4916	0.427	0.066	1.000	0.888	8.29	6	*	2.72	Classic OBP
5	RpadOBP3	0.4906	0.350	0.162	1.000	0.951	11.00	6	*	8.90	Classic OBP
6	GmolOBP7	0.4879	0.381	0.088	1.000	0.964	9.81	7		4.31	Classic OBP
7	SmosOBP12	0.4683	0.386	0.110	1.000	0.780	9.62	6	*	5.29	Classic OBP
8	HoblOBP2	0.4569	0.248	0.200	1.000	0.970	16.04	6	*	16.04	Classic OBP
9	PxylGOBP2	0.4496	0.437	0.025	0.850	0.795	7.97	6		0.98	GOBP2
10	RpadOBP7	0.4465	0.294	0.095	1.000	0.935	13.50	6	*	6.40	Classic OBP

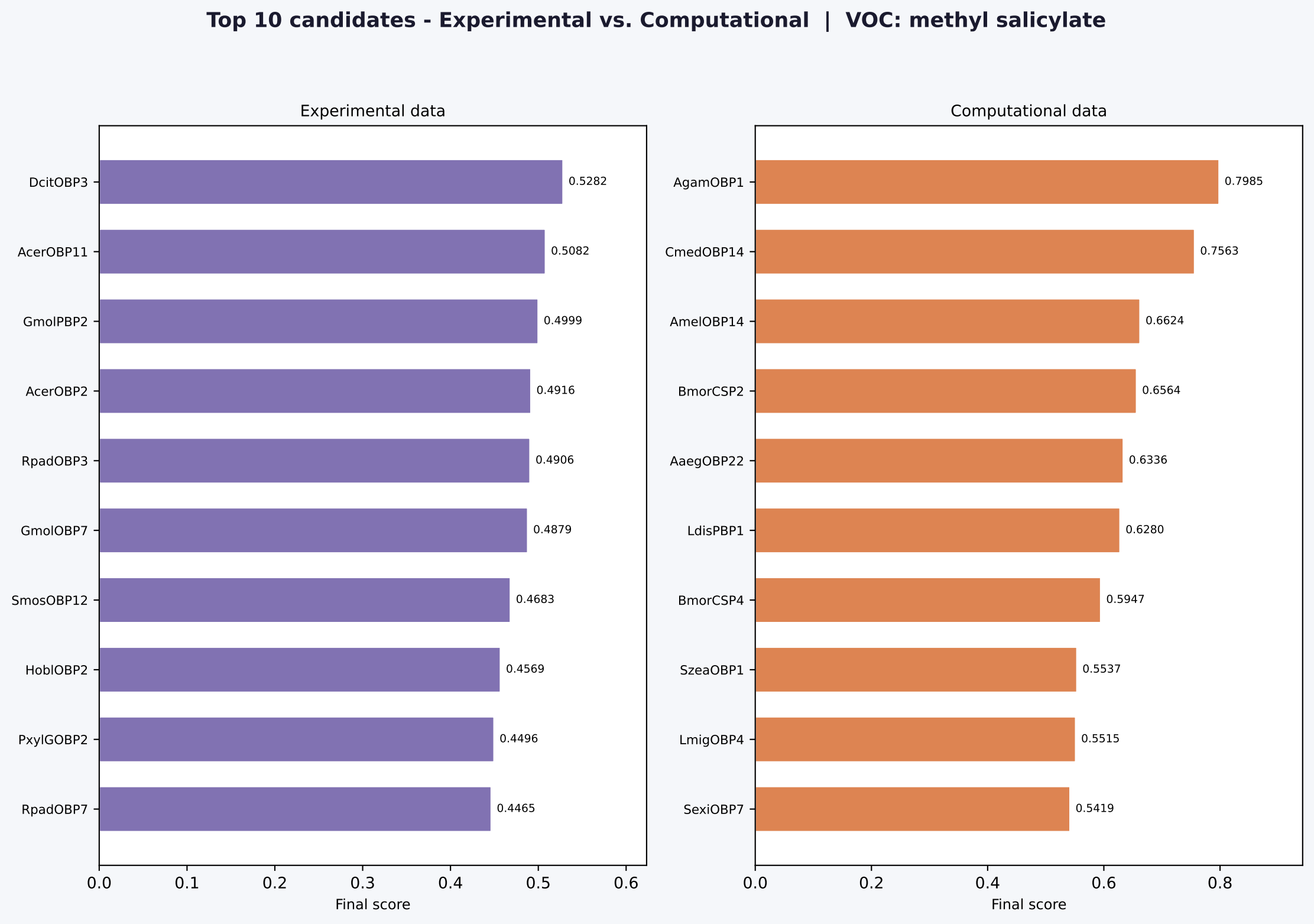

Top 10 candidates - Experimental vs. Computational | VOC: methyl salicylate
Experimental data
DcitOBP3
AcerOBP11
GmolPBP2
AcerOBP2
RpadOBP3
GmolOBP7
SmosOBP12
HoblOBP2
PxylGOBP2
RpadOBP7
0.5282
0.5082
0.4999
0.4916
0.4906
0.4879
0.4683
0.4569
0.4496
0.4465
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Final score
Computational data
AgamOBP1
CmedOBP14
AmelOBP14
BmorCSP2
AaegOBP22
LdisPBP1
BmorCSP4
SzeaOBP1
LmigOBP4
SexiOBP7
0.7985
0.7563
0.6624
0.6564
0.6336
0.6280
0.5947
0.5537
0.5515
0.5419
0.0
0.2
0.4
0.6
0.8
Final score

Table 1 / Table 2 - Reliable selectivity vs. promising | VOC: methyl salicylate

x No OBP has ALL interferents measured - selectivity is not reliable for any candidate.
Table 1 is not shown. Jumping straight to Table 2.

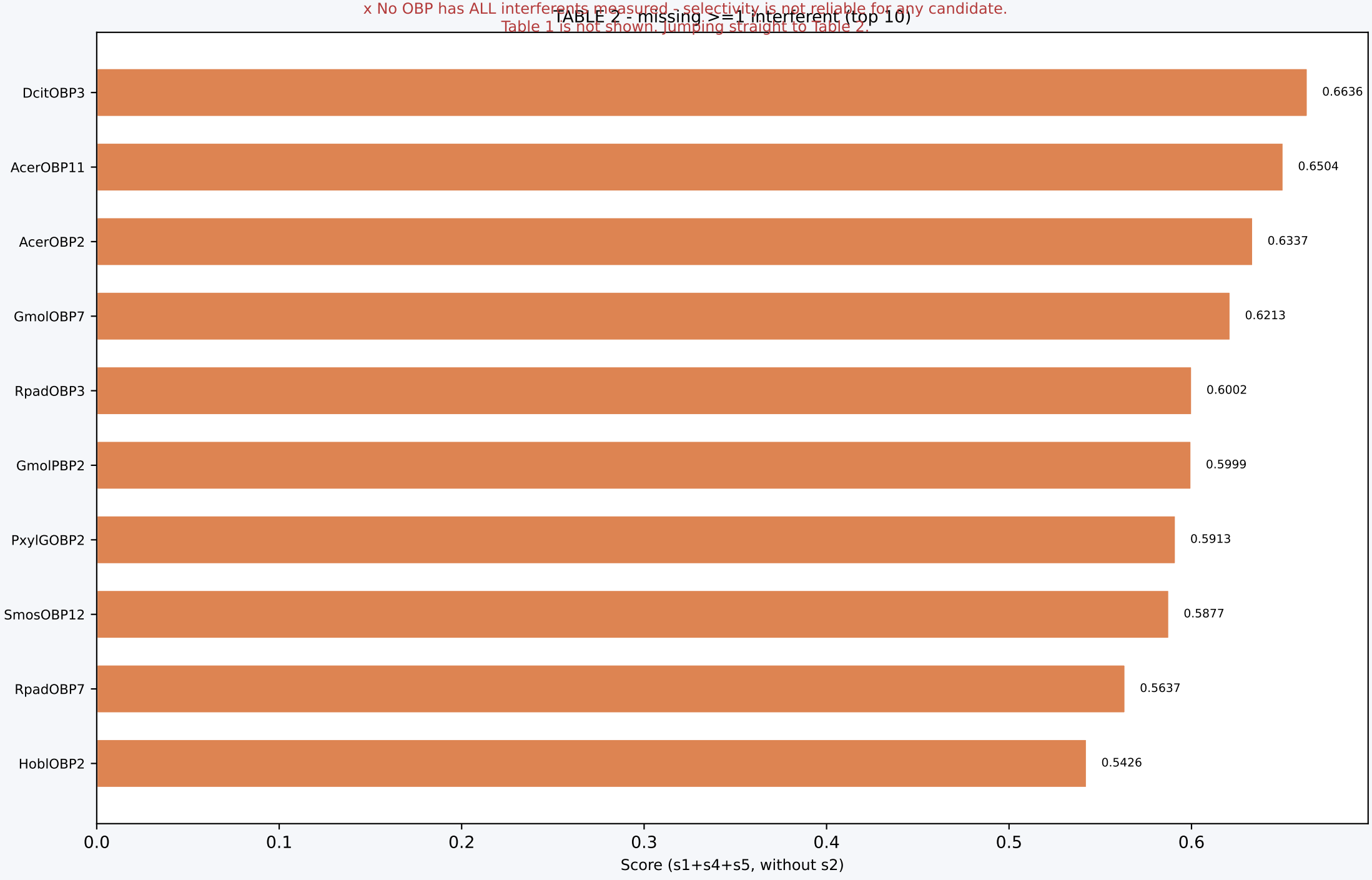
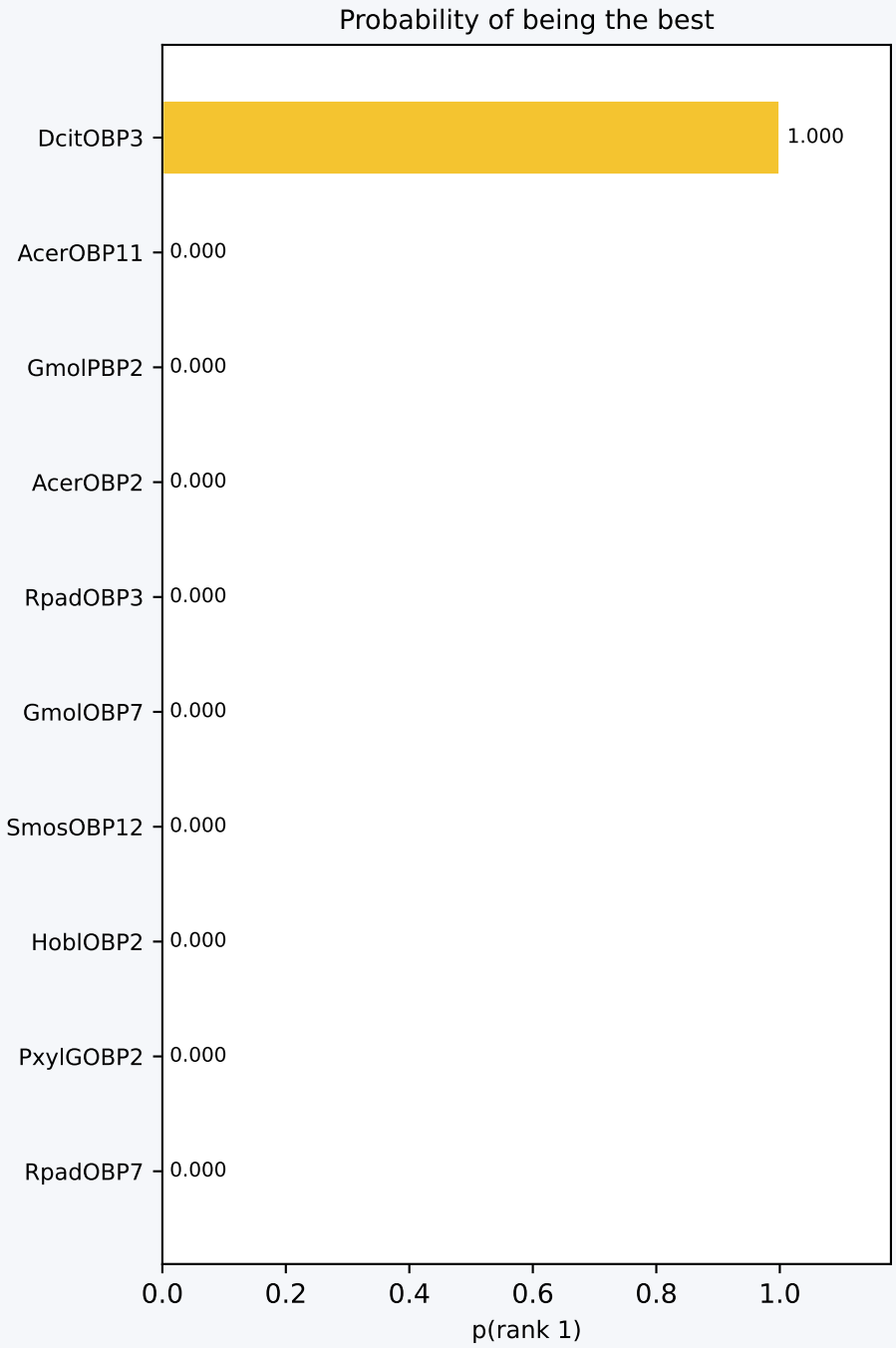


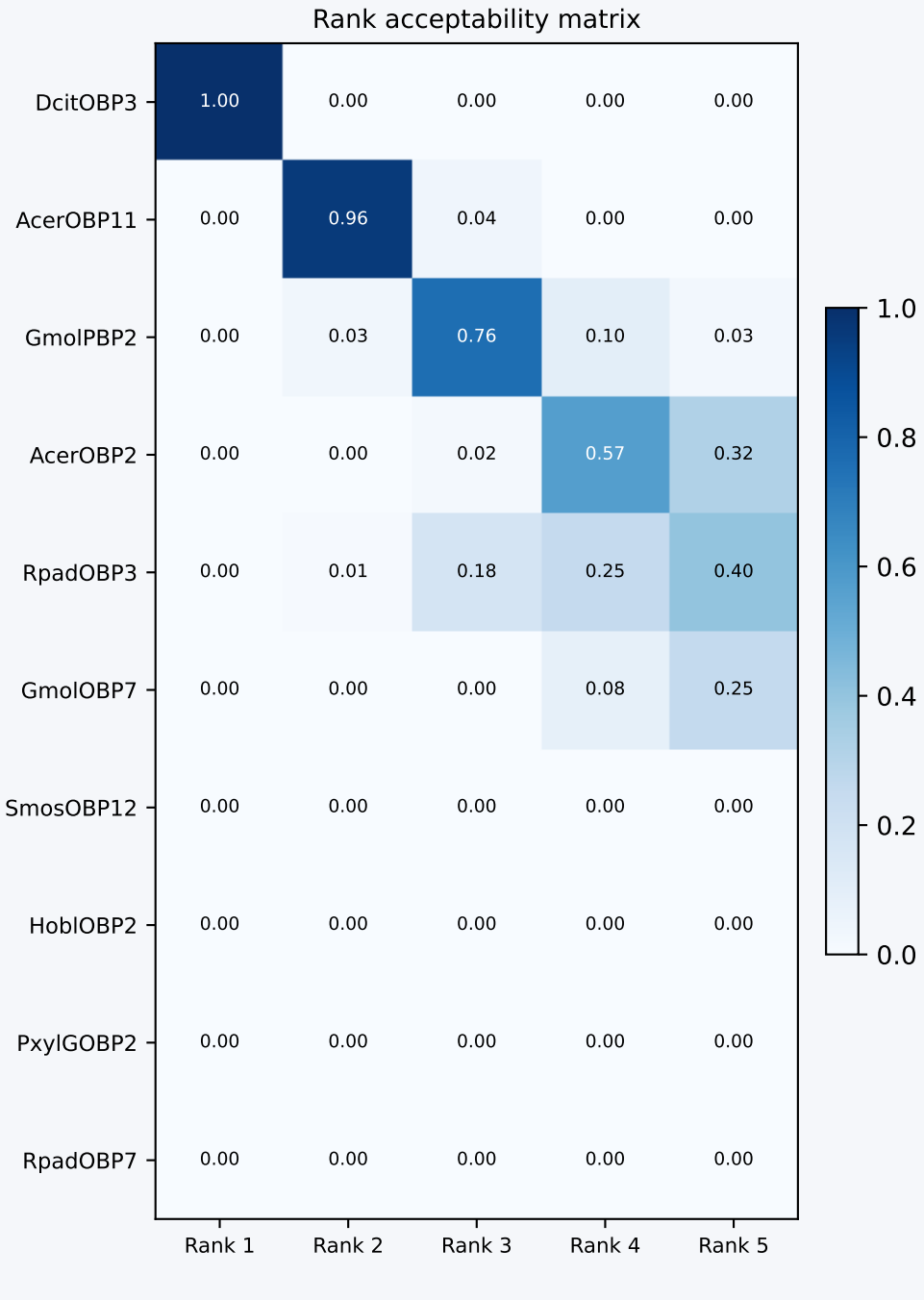
Table 3 - Boltz worklist | VOC: methyl salicylate

#	OBP	Table 2 Score	Missing interferent
1	DcitOBP3	0.6636	(+
2	DcitOBP3	0.6636	1-nonanol
3	DcitOBP3	0.6636	1-octanol
4	DcitOBP3	0.6636	1-octen-3-ol
5	DcitOBP3	0.6636	1-octen-3-one
6	DcitOBP3	0.6636	10:Ald
7	DcitOBP3	0.6636	12:Ald
8	DcitOBP3	0.6636	2-ethyl-1-hexanol
9	DcitOBP3	0.6636	2-hexen-1-ol
10	DcitOBP3	0.6636	2-pentylfuran
11	DcitOBP3	0.6636	2-phenyl ethanol
12	DcitOBP3	0.6636	2-undecanone
13	DcitOBP3	0.6636	3-octanone
14	DcitOBP3	0.6636	C8:Ald
15	DcitOBP3	0.6636	Ethyl hexadecanoate
16	DcitOBP3	0.6636	Farnesene
17	DcitOBP3	0.6636	Geranylacetone
18	DcitOBP3	0.6636	Ocimene
19	DcitOBP3	0.6636	Phellandrene
20	DcitOBP3	0.6636	alpha Terpineol
21	DcitOBP3	0.6636	benzaldehyde
22	DcitOBP3	0.6636	benzyl alcohol
23	DcitOBP3	0.6636	beta-farnesene
24	DcitOBP3	0.6636	caproic acid
25	DcitOBP3	0.6636	diisobutyl phthalate
26	DcitOBP3	0.6636	furfural
27	DcitOBP3	0.6636	geraniol
28	DcitOBP3	0.6636	hexanal
29	DcitOBP3	0.6636	nerol
30	DcitOBP3	0.6636	p-xylene
31	DcitOBP3	0.6636	phenylethanal
32	DcitOBP3	0.6636	stearic acid
33	DcitOBP3	0.6636	styrene
34	DcitOBP3	0.6636	α-Terpinene
35	DcitOBP3	0.6636	β-Cyclocitral

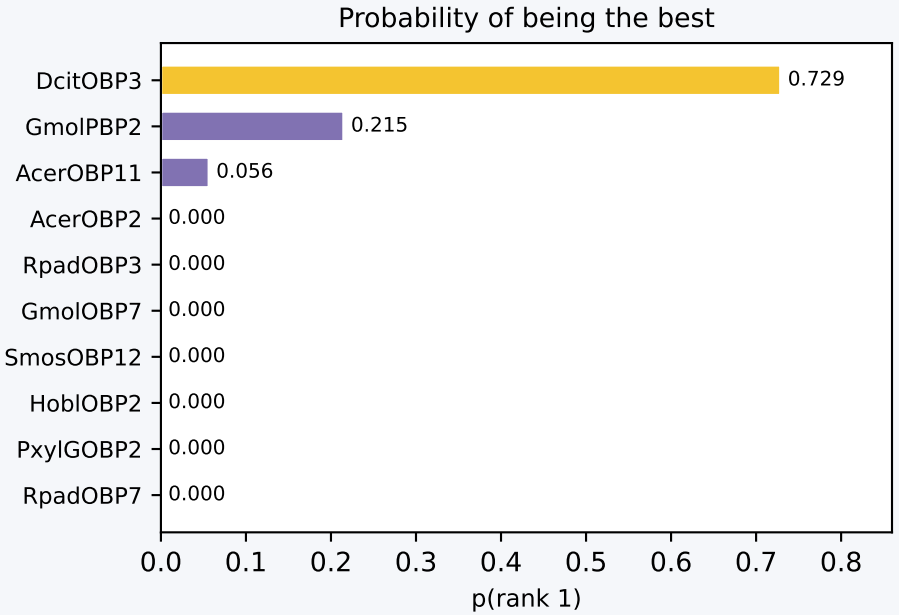
Total OBP/interferent pairs to compute: 3464 (showing the first 35, prioritized by Table 2)

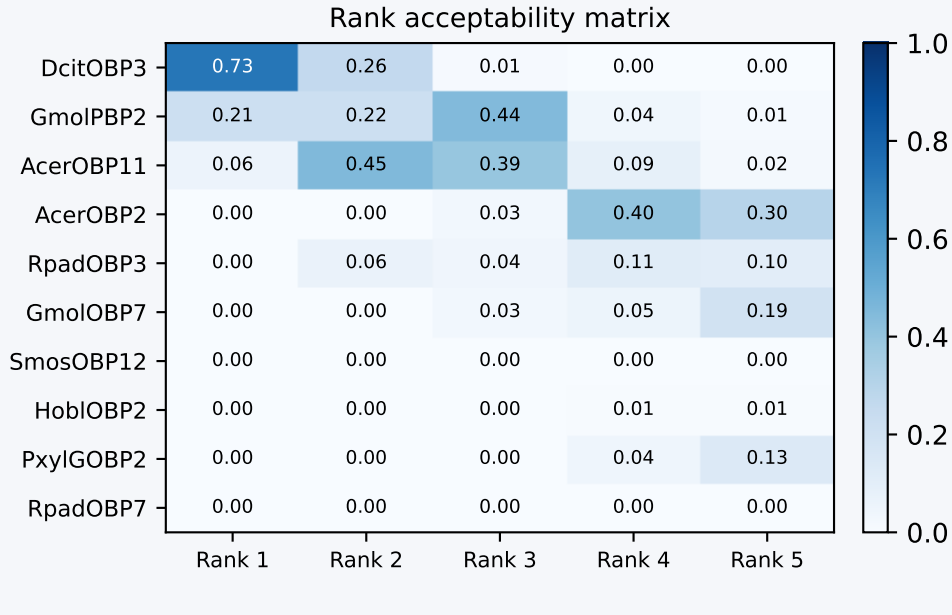
Method 2 - Weight Perturbation | VOC: methyl salicylate

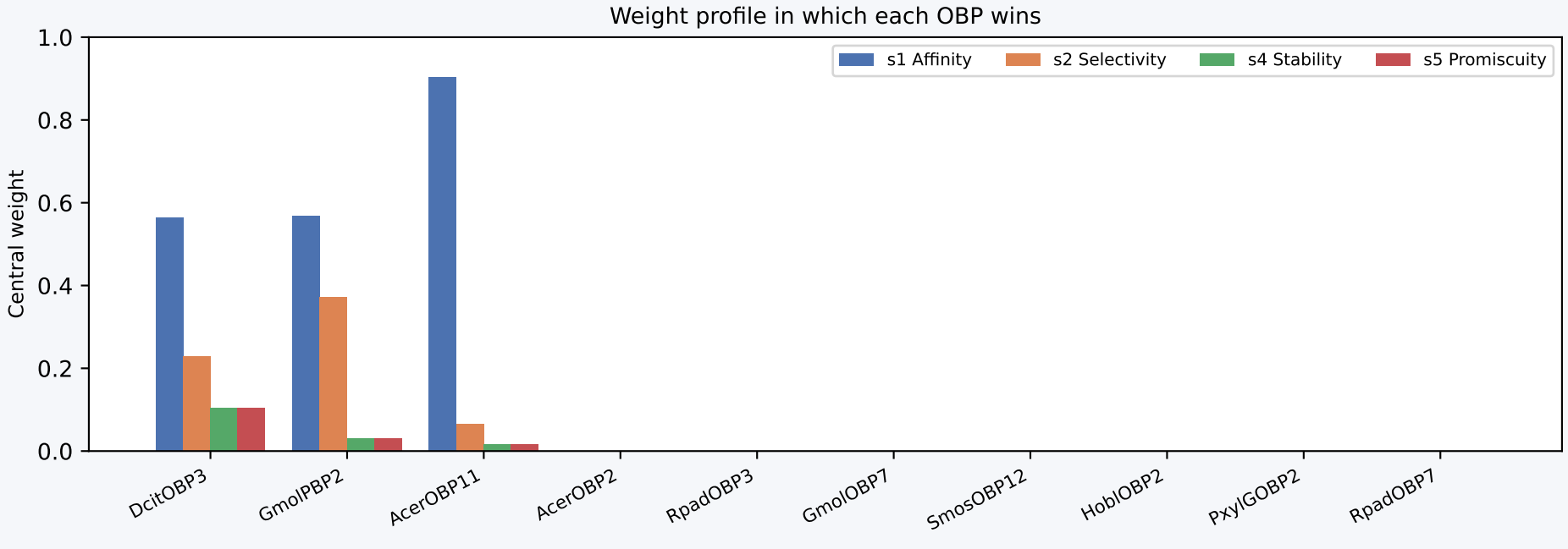


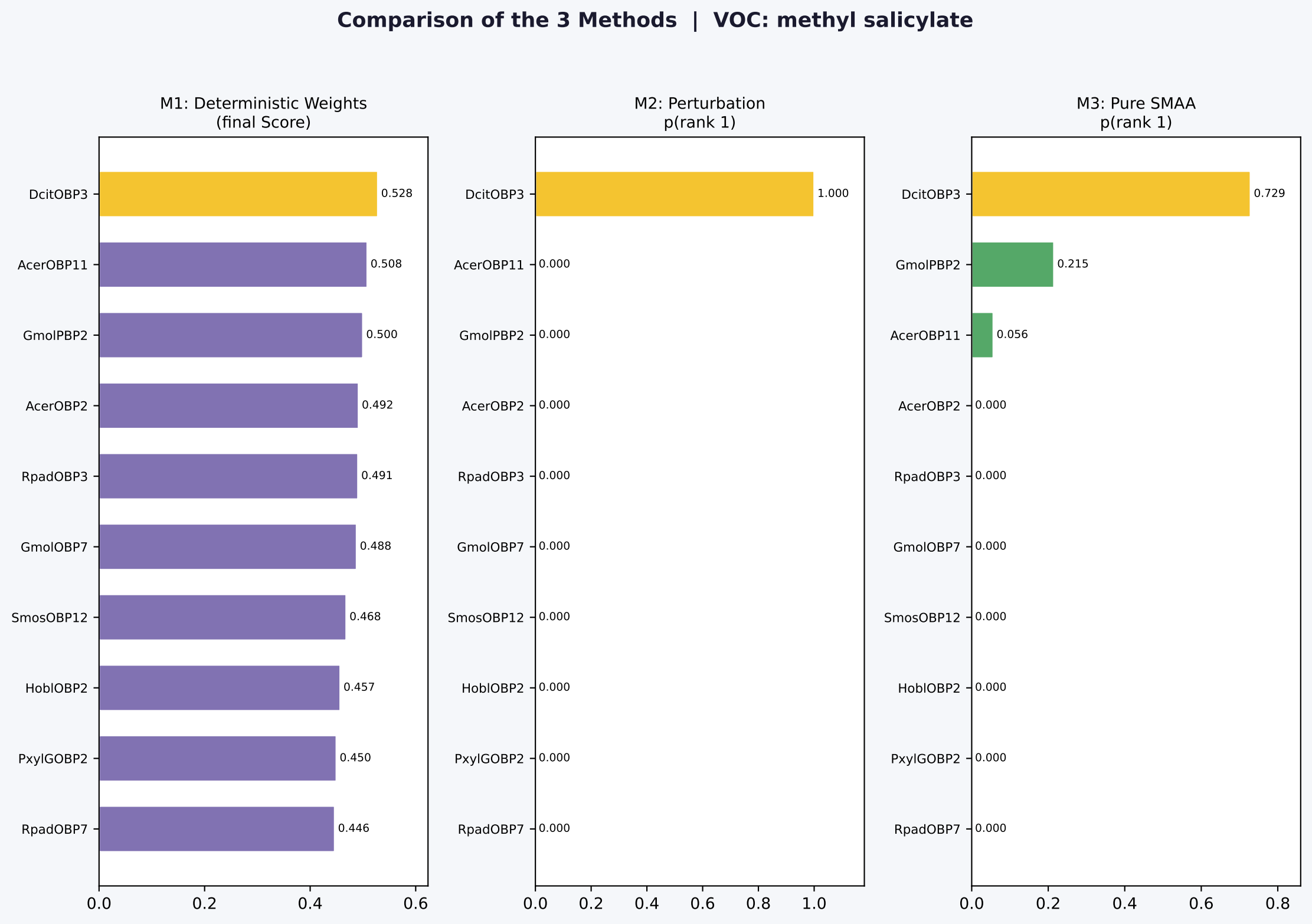


Method 3 - Pure SMAA | VOC: methyl salicylate








Comparison of the 3 Methods | VOC: methyl salicylate
M1: Deterministic Weights
(final Score)
DcitOBP3
AcerOBP11
GmolPBP2
AcerOBP2
RpadOBP3
GmolOBP7
SmosOBP12
HoblOBP2
PxylGOBP2
RpadOBP7
0.528
0.508
0.500
0.492
0.491
0.488
0.468
0.457
0.450
0.446
0.0
0.2
0.4
0.6
M2: Perturbation
p(rank 1)
DcitOBP3
AcerOBP11
GmolPBP2
AcerOBP2
RpadOBP3
GmolOBP7
SmosOBP12
HoblOBP2
PxylGOBP2
RpadOBP7
1.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.0
0.2
0.4
0.6
0.8
1.0
M3: Pure SMAA
p(rank 1)
DcitOBP3
GmolPBP2
AcerOBP11
AcerOBP2
RpadOBP3
GmolOBP7
SmosOBP12
HoblOBP2
PxylGOBP2
RpadOBP7
0.729
0.215
0.056
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.0
0.2
0.4
0.6
0.8